

Democratizing Data Analytics in Education: A Governance Strategy for Non-Profit Educational Organizations

Gabriel Verbel

*Facultad de Ingeniería de Sistemas Computacionales
Universidad Tecnológica de Panamá (UTP)
Panama City, Panama
gabriel.verbel@utp.ac.pa*

Kevin Gonzalez

*Facultad de Ingeniería de Sistemas Computacionales
Universidad Tecnológica de Panamá (UTP)
Panama City, Panama
kevin.gonzalez7@utp.ac.pa*

Ana Cecilia Muñoz

*Centro de Estudios en Microprocesadores y Sistemas Digitales (CEMISID)
Universidad de Los Andes
Mérida, Venezuela
acmunoz@ula.ve*

Victor Lopez

*Facultad de Ingeniería de Sistemas Computacionales
Universidad Tecnológica de Panamá (UTP)
Panama City, Panama
victor.lopez@utp.ac.pa*

Javier E. Sanchez-Galan

*Facultad de Ingeniería de Sistemas Computacionales
Universidad Tecnológica de Panamá (UTP)
Sistema Nacional de Investigación, SENACYT
Panama City, Panama
javier.sanchezgalan@utp.ac.pa*

Abstract—Implementing a data governance strategy is a complex task, particularly for non-profit organizations (NGOs) facing challenges such as limited resources and expertise gaps. This paper presents a case study on building a data engineering and governance framework for the Ayudinga Foundation, an educational NGO. The work begins with an initial data maturity assessment using the TDWI framework, which revealed a baseline 'Basic' literacy level. Based on this diagnosis, a comprehensive strategy was designed and implemented, including a Data Lakehouse architecture and a formal data governance program. The post-implementation assessment demonstrates a significant improvement, elevating the organization's overall data literacy maturity from 'Basic' (Level 3) to 'Literate' (Level 4). This paper details the methodology, technical implementation, and governance policies, offering a practical blueprint for similar organizations seeking to become data-driven.

Index Terms—Data Governance, Data Literacy, Data Lakehouse, Non-Profit Organization, TDWI Assessment

I. INTRODUCTION

In today's digital age, becoming a "data-driven organization" has become a fundamental strategic objective for institutions across all sectors [1]. This approach, based on using data to inform decisions, allows evaluation of results with concrete metrics and helps adjust initiatives based on verifiable evidence, generating greater impact and operational efficiency [2], [3].

The Ayudinga Foundation is a nonprofit digital educational organization in Panama dedicated to democratizing quality

education through accessible technological platforms. Since its inception, it has positively impacted thousands of students in Panama and Latin America through innovative educational content and academic support programs. However, to maximize its social impact, the foundation has identified the need to evolve toward a data-driven organizational structure.

The implementation of a big data infrastructure and a data governance framework is imperative for the foundation. Currently, data sources are distributed across multiple platforms and formats, limiting their potential to generate actionable insights. This fragmentation makes it difficult to accurately measure educational impact and hinders evidence-based decision making, essential elements for an educational organization seeking to maximize its reach and influence [4].

The main objective of this work is to report the implementation details of a comprehensive data governance strategy for the Ayudinga Foundation's big data environment. The overarching goal is to significantly increase the Foundation's data literacy maturity at the organizational level. To achieve this, an initial assessment was performed using the TDWI methodology [5], a known methodology that allows diagnosis of an organization's current data management maturity level. Based on these initial results, a centralized data infrastructure was designed and implemented, complemented by defined processes, rules, and roles that constitute the foundation's data governance framework.

This paper outlines the methodology used, details the tech-

nical architecture, presents the governance framework, and discusses the results and key findings.

II. RELATED WORK

A. Data Literacy and TDWI Assessment

Data literacy has become a critical component for organizations seeking to transform themselves into data-driven entities. According to Panetta et al. [6], "data literacy refers to the ability to read, write, and communicate data in context." Organizations increasingly recognize that this skill set is essential for candidates and employees in today's business environment.

To systematically assess and develop data literacy, models such as the TDWI Data Literacy Maturity Model offer structured frameworks that allow organizations to diagnose their current situation and plan improvements [7], [8]. Davenport and Bean [9] state that "organizations implementing AI are those that have previously developed a strong data culture and analytical skills throughout the organization."

B. Data Governance Frameworks

Data governance constitutes the procedural component of data-driven transformation. Talend defines data governance as "the process of managing the availability, usability, integrity, and security of an enterprise's data, based on internal standards and policies that also govern data usage" [10].

Profisee et al. identifies that an effective data governance framework should include well-defined roles, clear policies, and standardized processes [11]. In Ransbotham et al. [12], the authors found that data quality and governance processes are the main obstacles to successful AI/ML implementation.

Data governance refers to a series of processes, roles, policies, standards, and metrics to ensure the effective and efficient use of information that organizations use to achieve their objectives [13]. The strategy must be defined for each specific scenario, prioritizing the security of the most sensitive data in organizations that handle private customer data or allowing open access to their data in other organizations that handle publicly available data.

C. Data Architecture Evolution

Data architecture has undergone a clear transition over time, marked by three main technological generations: data warehouses, data lakes, and lakehouses. This evolution responds to the need to consolidate and analyze information from multiple operational systems, sensors, and unstructured sources, supporting decision-making in organizations [14].

The lakehouse architecture combines the best features of data lakes and data warehouses, providing structured data management capabilities while maintaining the flexibility to handle diverse data types and formats. This approach is particularly suitable for organizations like NGOs that need to manage varied data sources with limited technical resources.

III. METHODOLOGY

The implementation methodology was structured in three phases, each targeting specific aspects of the data governance transformation. As can be seen in Figure 1, this strategy included: a Phase 1, where an initial assessment of data management maturity was done. Phase 2 was needed to decode which data management solutions will be used. Finally, a phase 3, where the formal governance strategy was established.

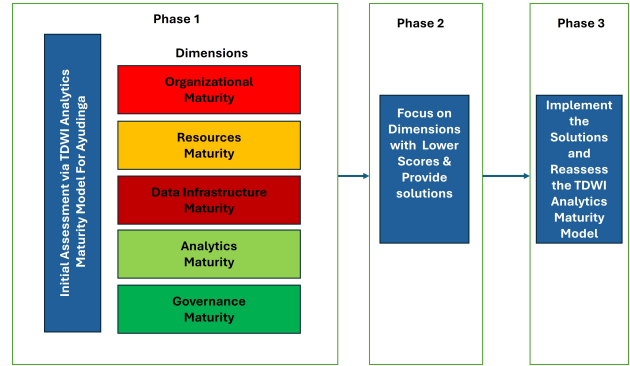


Fig. 1. Overview of the Three-Phase Data Governance Strategy Implementation Methodology

A. Phase 1: Initial Data Maturity Assessment

The project's first step was to understand the initial state of the Ayudinga Foundation's ability to manage and utilize data. The TDWI Data Literacy Maturity Assessment was applied, focusing on five key dimensions:

- 1) *Culture and Resources*: Measures the maturity of strategies, funding, and organizational support for data literacy.
- 2) *Data Infrastructure*: Evaluates the technologies and facilities for accessing and using data.
- 3) *Skills and Talent*: Analyzes the readiness of human talent to make data-driven decisions.
- 4) *Tools*: Considers the ease of adoption and diversification of tools for consuming and managing data.
- 5) *Data Governance and Processes*: Examines the processes implemented for orderly data control.

B. Phase 2: Implementation of Data Architecture

During Phase 2, the design and deployment of the data architecture (Lakehouse) was implemented using cloud-based solutions. The selected tool chain included open source and commercial components for ingestion, processing, and data delivery. Data models were defined to balance flexibility and performance, with storage formats and metadata layers ensuring discoverability and lineage tracking.

C. Phase 3: Establishment of Formal Governance Strategy

Phase 3 formalized the governance strategy by developing and documenting policies for data quality, privacy, and security. This included defining roles and responsibilities, creating a governance committee, and implementing processes for ongoing data quality monitoring and risk assessment.

IV. RESULTS

A. Initial Data Maturity Assessment

The initial TDWI assessment results placed the Ayudinga Foundation at a general stage 3 (basic literacy). Table I shows the detailed results across all dimensions.

TABLE I
INITIAL TDWI DATA LITERACY MATURITY ASSESSMENT RESULTS

Dimension	Stage	Level	Score
Culture and Resources	3	Basic	11
Data Infrastructure	2	Nascent	10
Skills and Talent	4	Advanced	15
Tools	2	Nascent	8
Data Governance	2	Nascent	9
Overall	3	Basic	-

The results revealed significant opportunities for improvement, especially in "Data Infrastructure" and "Data Governance and Processes" dimensions, which had the greatest potential impact for achieving a data-driven organization.

B. Data Architecture Implementation

The second phase focused on designing and implementing a robust, centralized data architecture conceived as a "Lakehouse." The architecture established a centralized location for all foundation data, allowing leverage by various business areas and facilitating self-management as a single source of truth.

1) *Architecture Design*: The functional requirements included:

- Centralized data repository
- Data input flows (ETL/ELT) for data updates
- Integration with various data navigation systems
- Orchestration system for automated process execution
- Data documentation mechanisms
- Support for semantic layers
- Data exploration environments

The proposed architecture, shown in Figure 2, materialized in a three-tier structure ("Data Producers," "Warehouse," and "Data Consumers") implemented on Huawei Cloud, using Dremio as the Lakehouse Engine.

2) *Technology Stack*: The implementation relied on:

- *Lakehouse Engine*: Dremio
- *Cloud Provider*: Huawei Cloud
- *Orchestration*: Kubernetes
- *Containerization*: Docker
- *Storage*: Elastic Volume Service (EVS)
- *Data Sources*: PostgreSQL

3) *Semantic Layers Implementation*: A semantic layers methodology was implemented, organizing Virtual Datasets (VDS) into folders within each Space:

- 1) *Staging*: VDS promoted from Physical Datasets with basic cleanup
- 2) *Features*: VDS with applied business logic and calculations

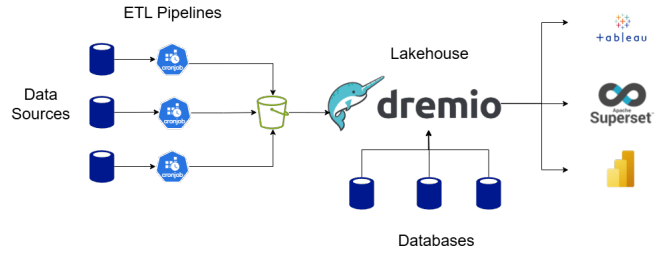


Fig. 2. Proposed Three-Tier Data Lakehouse Architecture

- 3) *Master Tables*: Consolidated master tables for analytics
- 4) *Results*: VDS storing analytics model results
- 5) *Business Intelligence*: Aggregated VDS for BI tool consumption

Figure 3 illustrates the developer workflow, which began with integrating new data sources into Dremio by establishing connections to PostgreSQL databases. This step made raw tables and views available for downstream processing and analysis. Next, the team organized the semantic layers by creating dedicated Spaces and folder structures.

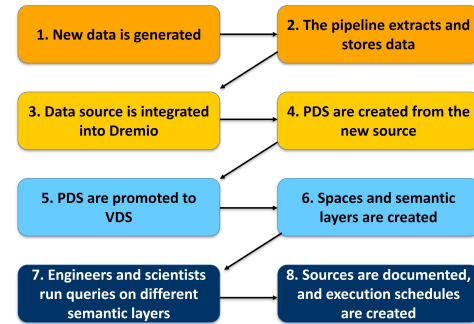


Fig. 3. End-to-End Workflow for Data Source Integration and Transformation

C. Data Governance Strategy

The third phase established a formal framework structured into four main components:

1) *Data Source Structuring*: Three hierarchical levels were established:

- *Attribute*: Specific element (column level)
- *Source*: Set of attributes with single purpose (table level)
- *Domain*: Grouping of sources by business theme

2) *Data Governance Roles*: Four key roles were defined:

- *Head of Data Governance*: Strategy implementation responsibility
- *Data Owner*: Responsible for data domain
- *Data Steward*: Expert in specific data sources
- *Data Worker*: Technical implementation and maintenance

The results of these access permissions can be seen in Table II.

TABLE II
ACCESS PERMISSIONS TO SEMANTIC LAYERS BY DATA GOVERNANCE
ROLE.

Data Governance Role	Semantic Layer	Permissions
Data Governance Lead	Business	R, D
	Intelligence	
	Result Table	R, D
	Master Table	R, D
	Feature	R, D
Data Owner	Staging	R, D
	Business	C, R, D
	Intelligence	
	Result Table	R
	Master Table	C, R, D
Data Steward	Feature	R
	Staging	R
	Business	C, R, D
	Intelligence	
	Result Table	R
Data Worker	Master Table	C, R, D
	Feature	C, R, U, D
	Staging	C, R, U, D
	Business	R, D
	Intelligence	

Note: Permissions are denoted as C = Create, R = Read, U = Update, and D = Delete.

3) *Data Governance Policies*: Six initial policies were established covering naming conventions, documentation requirements, glossary management, access permissions, quality metrics, and new data source implementation processes. Results can be found in Table III.

D. Post-Implementation Assessment

The post-implementation assessment, summarized in Table IV, confirmed that the project successfully met its objective. The targeted dimensions—Data Infrastructure, Tools, and Data Governance—all improved by two stages, elevating the foundation’s overall data literacy maturity from Stage 3 (Basic) to Stage 4 (Literate). While reaching Stage 5 was considered beyond the project’s initial scope, the implemented framework provides a solid foundation for future iterations and continuous improvement.

V. DISCUSSION AND CONCLUSIONS

This research demonstrates that structured data governance implementations can yield significant improvements in organizational data literacy, even within the resource constraints typical of NGO environments. The case study of Fundación Ayudinga provides several key findings with broader implications for data governance in non-profit contexts.

The research validates the effectiveness of the TDWI Data Literacy Maturity Assessment as both a diagnostic tool and an evaluation framework for governance implementations in

TABLE III
SUMMARY OF DATA GOVERNANCE POLICIES

Code	Policy Title	Description
P-01	Policy for Naming Domains, Sources and Attributes	Establishes standardized naming conventions (use of codes, snake_case) to ensure consistency and facilitate understanding.
P-02	Policy for Documentation Elements in Data Definitions and Catalogs	Details the minimum documentation elements required for each domain, source, and attribute across semantic layers.
P-03	Policy for Managing the Data Glossary	Defines the location of the glossary and processes for term inclusion (when creating new domains/sources or via user suggestions).
P-04	Policy for Data Access Permissions	Establishes the permissions (CREATE, READ, UPDATE, DELETE) for each data-governance role over the various Lakehouse semantic layers.
P-05	Policy for Selection and Implementation of Data Quality Metrics	Proposes an initial set of metrics (record counts, null values per column, unique fields) to evaluate source data quality.
P-06	Policy for Implementing New Data Sources	Describes a 7-step process for integrating new sources, from data-owner request to production deployment, including domain selection, initial input delivery, proof of concept (POC), pipeline implementation, documentation, access definition, and quality metric application.

TABLE IV
COMPARISON OF INITIAL AND FINAL MATURITY STAGES

Dimension	Initial	Final
Culture and Resources	3	3
Data Infrastructure	2	4
Skills and Talent	4	4
Tools	2	4
Data Governance	2	4
Overall	3	4

NGO settings. The successful implementation of a Data Strategy at the Ayudinga Foundation, materialized through a Lakehouse with Dremio and an initial Data Governance framework, has allowed the content department to begin developing its data-driven decision-making capabilities efficiently.

The TDWI assessment confirmed significant improvement in the foundation’s data-driven decision-making maturity, underscoring the project’s success. These advances have encouraged other areas of the organization to recognize the importance of data and consider integrating new sources into Lakehouse, demonstrating a positive and expansive impact.

is noteworthy that the dimension of “Culture and Resources” remained at Stage 3 (Basic) in the post-implementation evaluation. This outcome was anticipated, as this dimension is tied to deep-rooted organizational change that extends beyond the scope of a purely technical implementation. Although the project successfully advanced foundation’s technical capabilities, achieving greater maturity in culture requires a longer-term strategy focused on people and processes. The primary impediments were a nascent understanding of strategic value data at the leadership level, the absence of a dedicated data-centric role such as a Chief Data Officer (CDO), and the corresponding lack of targeted funding for data initiatives at the time of evaluation. This finding underscores a critical lesson: while technology is an enabler, fostering a data-driven culture is a separate, parallel effort that requires sustained leadership, strategic investment, and a focus on organizational change management [15].

The implemented strategy represents a first version with significant opportunities for improvement identified for future iterations. Among them is the need to address unstructured data sources, strengthen data governance policies, and utilize additional TDWI assessments such as “Data Management Maturity”, “Analytics Maturity” and “Data Strategy Maturity.”

In conclusion, this research contributes to the growing field of data governance in non-profit sectors by providing a structured implementation framework that can be adapted by similar organizations. The findings demonstrate that targeted governance policies and technical implementations can significantly enhance an organization’s data literacy maturity, even with resource constraints typical of NGO environments.

ACKNOWLEDGMENT

The authors express their gratitude for the administrative support provided by Universidad Tecnológica de Panamá and Ayudinga Foundation. Research activities by J.E.S.-G. are supported by the Sistema Nacional de Investigación (SNI) of SENACYT, Panama. This research received partial funding from the Consorcio Centroamericano de Computación de Alto Rendimiento-CoCeCAR (45-CR) initiative of the CSUCA-IDRC collaborative program.

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